

Behavioral Safety Assurance Module - (BSAM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

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Architecture Ecosystem: Structured Reality Standards™

Architecture Family: AI Governance Architecture (AIG®)

Operational Layer: AI Behavioral Governance Layer

Governed Space: Behavioral Safety Assurance

Category: Governance & Enforcement

Subcategory: AI Behavioral Safety Governance

Type: Behavioral Safety Standard Module

Parent Standard: Behavioral Safety Standard (BSS)

Version: 1.0

Status: Canonical · Open Module

Origin Date: 26 June 2026

Compatibility: OOF Methodology OS · AI Governance Architecture (AIG®) · Governance Architecture (GOA™) · Operational Reality Architecture (ORA™) · Cognitive Governance Intelligence Architecture (CLIA®) · Memory Governance Intelligence Architecture (MGIA™) · Accountability Governance Architecture (AGA™) · Autonomous Systems Governance Architecture (ASGA™)

AI-Readable: Yes

Authority: OOF

Protection: MIP™ — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition System

Canonical Definition

Behavioral Safety Assurance Module (BSAM) defines the structural conditions under which AI behavioral safety is continuously demonstrated, maintained, monitored, and governed to provide sustained confidence that AI behavior remains safe throughout the complete behavioral lifecycle.

BSAM governs behavioral safety assurance.

The module establishes the governance conditions required to ensure that behavioral safety remains a continuously observable, evidence-supported, independently verifiable, and governance-valid operational condition.

Validation confirms that safety exists.

Assurance confirms that safety continues.

BSAM governs that assurance.

Module Operational Role

BSAM defines the behavioral-safety-assurance layer of BSS.

Its role is to provide continuous governance assurance that AI behavior remains safe during ongoing operation.

Module Operational Space

BSAM governs:

- behavioral safety assurance
- continuous safety assurance
- governance safety assurance
- operational safety confidence
- behavioral protection assurance
- safety monitoring assurance
- stakeholder safety confidence
- long-term safety governance

The module applies wherever organizations require continuous confidence that AI behavioral safety remains preserved over time.

Module Function

The module applies wherever systems must preserve:

- continuous behavioral safety,
- governance-supported assurance,
- evidence-based confidence,
- accountable safety oversight,
- operational trust,
- sustainable behavioral protection.

Its function is to ensure that behavioral safety remains continuously demonstrable rather than periodically confirmed.

Minimum Implementation Framework

1. Define the Behavioral Safety Assurance Object

The organization must define which AI behaviors require continuous behavioral safety assurance.

2. Define Assurance Conditions

The system must define governance conditions including:

- behavioral predictability,

- operational reliability,
- governance compliance,
- continuous monitoring,
- accountability,
- evidence availability.

3. Define Assurance Failure Detection Logic

The system must identify:

- declining safety assurance,
- emerging behavioral uncertainty,
- governance deficiencies,
- operational instability,
- reduced confidence,
- governance-invalid safety assurance.

4. Define Operational Response or Governance Logic

Governance response may include:

- assurance review,
- governance reassessment,
- enhanced monitoring,
- preventive intervention,
- operational restriction,
- continuous improvement.

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- assurance activities,
- governance reviews,
- behavioral observations,
- safety evidence,
- intervention decisions,
- resulting governance outcomes.

An AI behavioral environment must not be considered continuously safe unless behavioral safety assurance can be independently demonstrated, reviewed, validated, preserved, and continuously governed.

Use Case 1 — Autonomous Aviation Network

Scenario

An international aviation operator deploys AI to support autonomous flight

coordination across multiple countries.

Application

BSAM continuously assures that AI behavioral safety remains preserved despite changing operational conditions, traffic density, and environmental complexity.

Result

The aviation operator strengthens regulatory confidence, operational resilience, and long-term public trust in autonomous aviation.

Use Case 2 — National Healthcare AI Ecosystem

Scenario

A nationwide healthcare network relies on AI to support diagnostics, treatment planning, and hospital coordination.

Application

BSAM continuously demonstrates that behavioral safety remains governance-valid across thousands of daily AI-assisted clinical interactions.

Result

The healthcare system strengthens patient protection, governance maturity, and sustained confidence in safe AI-assisted healthcare.

Canonical Closing Statement

Safety is the final promise of responsible AI governance. Behavioral Safety Assurance ensures that this promise is continuously demonstrated through governed behavior, allowing people, organizations, and society to rely on AI with enduring confidence.

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Canonical Source

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